

Traffic splitting for safe

deployments: A useful feature is traffic splitting. Operators can slowly send a small amount of traffic to a new version while keeping an eye on performance and error rates. If there are problems, traffic can be quickly redirected so that users don't have to deal with much trouble.

SLOs and dependability: To keep things running smoothly in production environments, you need to have clear service level objectives (SLOs). When teams use Knative on top of Kubernetes, they need to keep an eye on performance metrics and set limits on how the system can behave.

Measuring cold start latency: Cold start latency is an important measure of reliability. The first request after a period of inactivity may take longer because Knative can scale services down to zero instances while a new container instance starts.

Setting performance budgets: Performance budgets set limits on how long it takes to respond, how many errors can happen, and how often the system is available. Teams can check if their serverless workloads are meeting operational goals by setting clear budgets.

The Horizontal Pod Autoscaler (HPA) in Kubernetes already lets you scale by changing the number of pods based on resource metrics like CPU usage. Knative has its own autoscaling system that works best with workloads that are based on requests.

Workflows for CI/CD and GitOps

To use Knative services in production, you need good version control and automation. Declarative YAML manifests are used to define most Knative deployments. These manifests describe services, revisions, and event resources.

Delivery that gets better over time: Knative's traffic management tools make it possible to use

progressive delivery strategies. Canary deployments and blue-green releases are two ways that teams can slowly roll out new versions of services while keeping an eye on how the system works

Standards for pod security: One important way to protect yourself is to follow Pod Security Standards (PSS). These rules limit what containers can do inside the cluster, such as giving them host access, root-level execution, and unsafe capabilities.

Signing images: You should always check container images before you deploy them. Image signing tools make sure that images come from reliable sources and haven't been changed.

Policies for the network: Network policies give you very precise control over how services talk to each other within the cluster. These rules limit traffic between services or namespaces, making sure that only authorised communication paths are allowed.

Safety in the supply chain:

Software supply chain security is becoming increasingly important because modern applications depend heavily on things outside of themselves. Before deployment, automated scanning tools can find flaws in container images and dependencies.

Risks of multi-tenant isolation: When multiple teams or organisations share a cluster, multi-tenant isolation becomes a big problem. Weak isolation policies can let workloads from one tenant get in the way of or use resources that belong to another tenant.

Trade-offs and performance benchmarks

When looking at serverless platforms,

companies often compare open source options like Knative with managed services like AWS Lambda. Each method has pros and cons when it comes to performance, scalability, and cost of doing business.

Managed serverless platforms usually offer a very optimised environment where the cloud provider takes care of all the infrastructure management. This lets developers only work on the code for the application. Knative, on the other hand, offers a similar serverless experience but runs on Kubernetes clusters that businesses manage themselves.

Characteristics of scaling throughput

Knative works well in places where you need a lot of throughput and the ability to scale up or down as needed. It can scale horizontally by starting more container instances on different cluster nodes because it runs on Kubernetes.

Modelling costs: The structures of the costs are very different. Managed serverless platforms charge based on how many times a function is called and how long it takes to run, which is great for workloads that don't happen all the time.

Easy workload: If you need a small application or a simple background job, using a full Kubernetes-based serverless platform might make things more complicated than they need to be. Managed services like AWS Lambda can often give you results faster with little to no setup.

Knative works best in companies that are willing to spend money on engineering teams that take care of the infrastructure and tools for developers. **END** 

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